

John Lawrence Palacios

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TECHNICAL SKILLS

Modeling and Simulation: Physiologically-Based Pharmacokinetics (PBPK), Pharmacokinetics/Pharmacodynamics (PK/PD), Monte Carlo simulations, Sensitivity Analysis, Optimization
Programming and Software: R (mrgsolve, ospsuite), PK-Sim, MoBi, MATLAB, LaTeX

EDUCATION

Virginia Commonwealth University Richmond, VA
Doctor of Philosophy in Systems Modeling & Analysis Aug. 2020 – Aug. 2024
Dissertation: Mathematical Modeling of Phage-Bacteria Population Dynamics
University of Guam Mangilao, GU
Bachelor of Science in Mathematics Aug. 2013 – May 2018

EXPERIENCE

Oak Ridge Institute for Science and Education (ORISE) Fellow Aug. 2024 – Present
U.S. Food & Drug Administration Silver Spring, MD

- Developed and validated whole-body PBPK models for predicting antidepressant pharmacokinetics in healthy and pregnant populations
- Performed parameter estimation and sensitivity analysis to refine PBPK and PK/PD model predictions
- Curated and integrated in vitro and in vivo data from literature to enhance model validation
- Collaborated closely with a team of experienced modelers, pharmacologists, and physicians contributing to the iterative modeling process
- Utilized in silico cotyledon perfusion model to validate placental transporter parameters using ex vivo experimental data

Graduate Research Assistant Aug. 2020 – Aug. 2024
Virginia Commonwealth University Richmond, VA

- Optimized differential equation models to accurately fit in vitro experimental data
- Performed local and global sensitivity analysis to identify key model parameters
- Adapted an agent-based model to analyze spatial dynamics of biological systems such as biofilms
- Worked with cross-disciplinary teams to model phage-bacteria interactions

Graduate Teaching Assistant Aug. 2020 – June 2024
Virginia Commonwealth University Richmond, VA

- Guided students in applying mathematical techniques to real-world modeling problems
- Mentored students in independent research projects on mathematical modeling such as infectious disease spread

PUBLICATIONS

- J. Palacios**, R. Segal, “Exploration of Differential Equation Models for Phage-Bacteria Population Dynamics,” *Spora: A Journal of Biomathematics*, Vol. 11, 36-47, Feb. 2025, <https://doi.org/10.61403/2473-5493.1098>
- V.N. Campo, **J. L. Palacios**, H. Nagahashi, H. Oh, J. Rychtar, and D. Taylor, “A game-theoretic model of rabies in domestic dogs with multiple voluntary preventive measures,” *Journal of Mathematical Biology*, vol. 85, no. 5, Oct. 2022, doi: <https://doi.org/10.1007/s00285-022-01826-z>
- P. Gandhi, M. A. Robert, **J. Palacios**, D. Chan, “Effects of Contact Tracing and self-Reporting in a Network Disease Model,” *Letters in Biomathematics*, 2023, doi: <https://doi.org/10.30707/lib9.1.1681913305.219107>